

LESSON 17 / 第十七課

Tree Hydraulics, Drought and Irrigation: From Climate to the Vulnerability Curve

樹木水力、乾旱與灌溉：從氣候到脆弱性曲線

Format. Short literature review. Claims that carry a number or a causal ranking are tied to a numbered source. Scale runs from the largest published pattern to the finest measured process. The lesson is not sequenced from the previous number and is not a regional case file unless the object itself is regional.

體例。短文獻綜述。帶數字或因果排序的陳述繫到編號文獻。尺度由最大已發表格局寫到最細已測過程。本課不由上一號遞推；除非對象本身是地域性的，否則不是單一地區檔案。

Abstract

摘要

Water moves through trees under tension. The published control variable is water potential, not a percentage of soil moisture by folklore. Vulnerability curves report the potential at which conductivity is lost (P50, P88). Choat et al. showed that most forest species operate with safety margins narrower than 1 MPa, largely independent of mean annual precipitation. Isohydic and anisohydric strategies describe stomatal behaviour, not moral character. Urban pits multiply drought by volume restriction, heat and vapour-pressure deficit. An irrigation specification names a trigger, a dose, a shut-off and a term. Mulch is a surface specification, not a volcano against the stem.

水在張力下穿過樹木。已發表的控制變量是水勢，不是傳聞中的土壤水分百分比。脆弱性曲線報告導水率喪失時的水勢（P50、P88）。Choat 等顯示多數森林樹種的安全邊際窄於 1 MPa，且大致獨立於年均降水。等水與非等水策略描述氣孔行為，不是道德性格。城市樹穴以體積限制、高溫與水汽壓差倍增乾旱。灌溉規格點名觸發、劑量、關停與期限。覆蓋物是表面規格，不是抵住莖的火山。

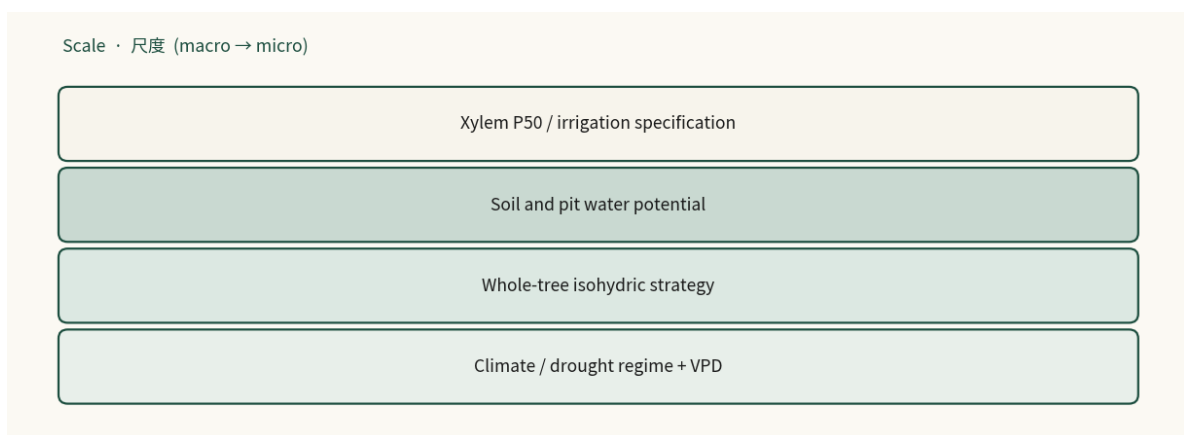


Figure 1. Nested scales from climate to the vulnerability curve.

圖 1. 由氣候到脆弱性曲線的嵌套尺度。

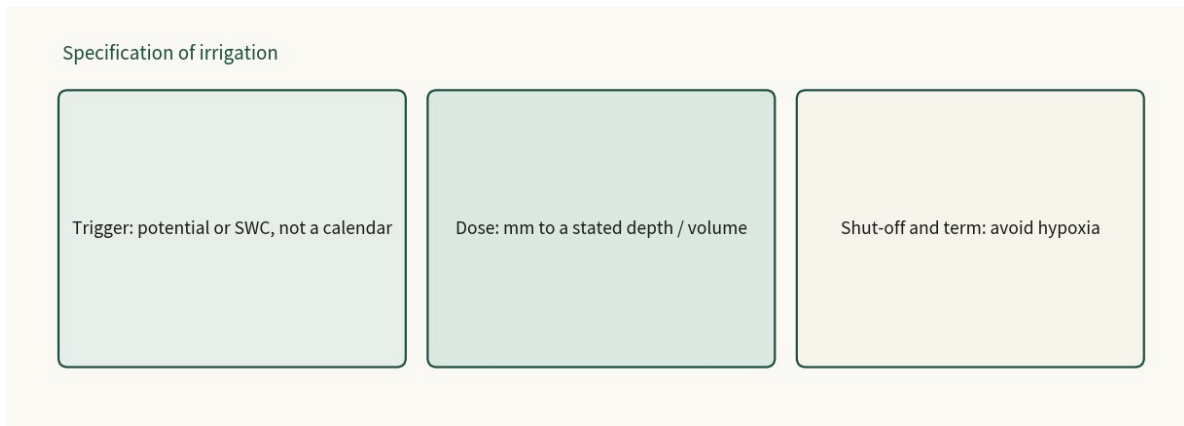


Figure 2. Irrigation as trigger, dose, shut-off and term.

圖2. 灌溉作為觸發、劑量、關停與期限。

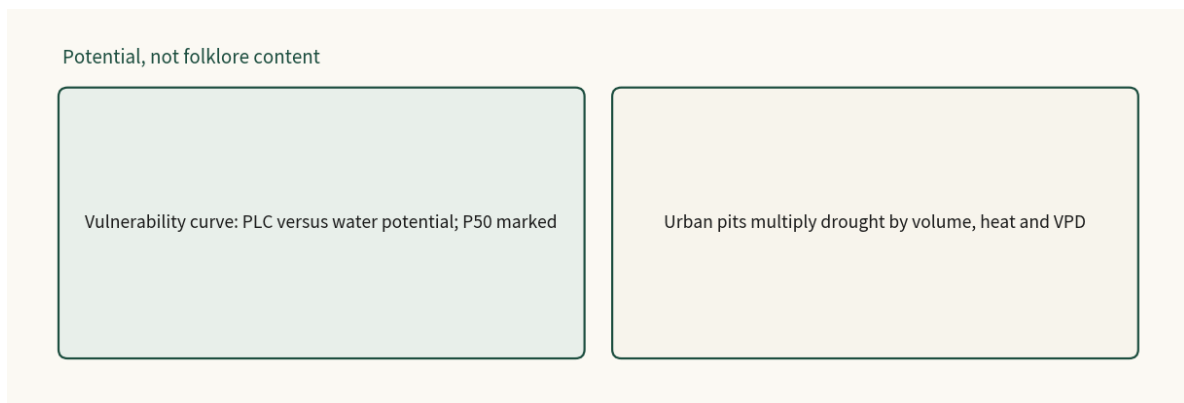


Figure 3. Water potential, not a folklore moisture percentage.

圖3. 水勢，不是傳聞中的水分百分比。

1. Scope

1. 範圍

The object is published tree hydraulics as it constrains urban drought and irrigation specification. Mortality mechanisms at death are Lesson 24; this review stops at the curves and the specification.

對象是約束城市乾旱與灌溉規格的已發表樹木水力。死亡當下的機制是第二十四課；本綜述停在曲線與規格。

2. Potential, curves, strategies

2. 水勢、曲線、策略

Cohesion-tension is the published ascent mechanism [1,2]. A vulnerability curve plots percent loss of conductivity against xylem water potential; P50 is the potential at 50% loss [3]. Choat et al. compiled 226 species at 81 sites and found that 70% operate with hydraulic safety margins narrower than 1 MPa, largely independent of mean annual precipitation [4]. Isohydric taxa close stomata to defend potential; anisohydric taxa allow potential to fall further — a behaviour spectrum, not a binary essence [5]. Savi and colleagues documented hydraulic deterioration on urban trees as a chronic, not only acute, process [6].

內聚–張力是已發表的上升機制 [1,2]。脆弱性曲線把導水率損失百分比對木質部水勢作圖；P50 是損失 50% 時的水勢 [3]。Choat 等彙整 81 個樣點、226 個樹種，發現 70% 的水力安全邊際窄於 1 MPa，且大致獨立於年均降水 [4]。等水類群關閉氣孔以保衛水勢；非等水類群容許水勢掉得更低

——這是行為譜，不是二元本質 [5]。Savi 及同事把城市樹上的水力惡化記錄為慢性過程，而不只是急性過程 [6]。

3. Urban multipliers and irrigation

3. 城市倍增因子與灌溉

Restricted volume, sealed heat and high vapour-pressure deficit multiply the same curve [6,7]. Irrigation specified only as ‘water regularly’ has no trigger, no dose, no shut-off and no term. The published specification names those four. Mulch depth and trunk clearance are A300 Part 2 items; volcano mulch is the opposite of that specification [8].

受限體積、封蓋熱與高水汽壓差倍增同一條曲線 [6,7]。只寫成「定期澆水」的灌溉沒有觸發、劑量、關停與期限。已發表規格點名這四項。覆蓋厚度與離幹間距是 A300 第 2 部分項目；火山狀覆蓋是該規格的反面 [8]。

4. Limits

4. 限度

Vulnerability-curve method has been revised (centrifuge, optical, pneumatic). The existence of a PLC–potential relationship is not in dispute; particular P50 numbers can move with method. Transfer of a forest P50 onto a paved pit without a volume and VPD term is incomplete.

脆弱性曲線方法已被修訂（離心、光學、氣壓）。PLC 與水勢關係的存在沒有爭議；特定 P50 數字可隨方法移動。沒有體積與水汽壓差項就把森林 P50 轉到鋪面樹穴，是不完整的。

Study questions

研習題

Choose the option that retains the published result. Answers follow the last item.

選出保留已發表結果的選項。答案在最後一題之後。

1. The published control variable is:

1. 已發表的控制變量是：

A. A folklore moisture percent.

A. 傳聞水分百分比。

B. Water potential.

B. 水勢。

C. t/R .

C. t/R 。

D. Canopy percent.

D. 樹冠百分比。

2. P50 is:

2. P50 是：

A. A soil texture.

A. 土壤質地。

B. The potential at 50% loss of conductivity.

B. 導水率損失 50% 時的水勢。

C. A lightning path.

C. 雷電路徑。

D. Wall 4.

D. 第 4 壁。

3. Choat et al. found safety margins:

3. Choat 等發現安全邊際：

A. Always wider than 3 MPa.

A. 總寬於 3 MPa。

B. Narrower than 1 MPa in 70% of species, largely independent of MAP.

B. 70% 樹種窄於 1 MPa，大致獨立於年均降水。

C. Defined only for palms.

C. 只為棕櫚定義。

D. Equal to 3-30-300.

D. 等於 3-30-300。

4. Isohydic / anisohydic describes:

4. 等水／非等水描述：

A. Moral character.

A. 道德性格。

B. Stomatal behaviour along a spectrum.

B. 譜上的氣孔行為。

C. Cable grades.

C. 鋼纜等級。

D. Register status.

D. 名錄身份。

5. An irrigation specification names:

5. 灌溉規格點名：

A. Only 'water regularly'.

A. 只有「定期澆水」。

B. Trigger, dose, shut-off and term.

B. 觸發、劑量、關停與期限。

C. Only flower colour.

C. 只有花色。

D. Only i-Tree.

D. 只有 i-Tree。

6. Urban pits multiply drought by:

6. 城市樹穴倍增乾旱，透過：

A. Nothing.

A. 無事。

B. Volume restriction, heat and VPD.

B. 體積限制、熱與水汽壓差。

C. Extra mycorrhizae only.

C. 只有額外菌根。

D. Flash density.

D. 閃擊密度。

7. Volcano mulch is:

7. 火山狀覆蓋是：

A. The A300 Part 2 specification.

A. A300 第 2 部分規格。

B. The opposite of that specification.

B. 該規格的反面。

C. A P88.

C. 一個 P88。

D. A tomogram.

D. 斷層圖。

8. Savi et al. treated urban hydraulic deterioration as:

8. Savi 等把城市水力惡化視為：

A. Only acute.

A. 只有急性。

B. Chronic as well as acute.

B. 慢性也有。

C. Impossible.

C. 不可能。

D. A diversity rule.

D. 多樣性規則。

9. Particular P50 numbers:

9. 特定 P50 數字：

A. Never move with method.

A. 永不隨方法移動。

B. Can move with method; the PLC-potential relation remains.

B. 可隨方法移動；PLC 與水勢關係仍在。

C. Equal occupancy.

C. 等於佔用。

D. Set SOS.

D. 設定 SOS。

10. Transfer of a forest P50 onto a paved pit without volume and VPD:

10. 沒有體積與水汽壓差就把森林 P50 轉到鋪面樹穴：

A. Is complete.

A. 完整。

B. Is incomplete.

B. 不完整。

C. Is Wall 1.

C. 是第 1 壁。

D. Is QTRA.

D. 是 QTRA。

Answers / 答案 1-B 2-B 3-B 4-B 5-B 6-B 7-B 8-B 9-B 10-B

References

參考文獻

- [1] Zimmermann M.H. (1983). Xylem Structure and the Ascent of Sap.
- [2] Tyree M.T., Zimmermann M.H. (2002). 2nd ed.
- [3] Sperry J.S., Tyree M.T. vulnerability-curve foundations.
- [4] Choat B. et al. (2012). Nature 491: 752-755.
- [5] Tardieu F., Simonneau T. isohydric / anisohydric behaviour.
- [6] Savi T. et al. (2015). hydraulic deterioration of urban trees.
- [7] Jim C.Y. urban-pit water and heat regimes.
- [8] ANSI A300 Part 2 soils / mulch.

Closing note. This lesson is written as a short review article. Disagreement in the literature is kept in view where the results actually diverge.

結語。本課以短綜述文章寫成。文獻真正分歧之處，兩造結果都保留在視野內。